

FORT ST JOHN, BC, Canada

WMO: 719430

Lat: 56.2467N

Lon: 120.7371W

Elev: 2280

StdP: 13.52

Time Zone: -7.00 (W07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -25.6	(c) -19.8	(d) -34.9	(e) 0.8	(f) -24.7	(g) -29.3	(h) 1.2	(i) -19.8	(j) 29.9	(k) 30.8	(l) 26.8	(m) 33.2	(n) 4.5	(o) 350	(p) 0.713

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.3	79.2	61.0	76.1	59.5	73.3	58.0	63.4	75.3	61.3	73.1	59.6	70.5	8.6	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
58.8	80.6	67.4	56.7	74.5	65.1	54.9	69.7	63.7	30.0	75.4	28.4	73.4	27.1	70.8	71.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
23.3	21.0	18.6	-30.7	85.2	5.9	2.9	-34.9	87.3	-38.4	89.0	-41.7	90.7	-46.0	92.8		
			-30.9	66.2	5.8	2.6	-35.1	68.1	-38.5	69.6	-41.8	71.1	-46.0	73.0		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	36.2	9.2	14.6	22.1	38.0	49.9	57.4	61.3	59.0	50.7	38.1	20.9	11.6
	DBStd	22.14	18.39	15.20	14.77	9.35	8.23	5.62	5.38	6.20	7.83	9.18	13.63	16.37
	HDD50	6135	1266	991	864	368	102	8	2	7	85	377	873	1192
	HDD65	10570	1731	1411	1329	811	470	235	138	202	430	833	1323	1657
	CDD50	1085	0	0	0	6	98	229	352	285	106	9	0	0
	CDD65	46	0	0	0	0	1	6	23	15	1	0	0	0
	CDH74	528	0	0	0	1	25	77	231	167	26	1	0	0
	CDH80	75	0	0	0	0	1	8	39	25	2	0	0	0
Wind	WSAvg	8.7	8.9	8.4	8.9	9.4	9.0	8.3	8.0	7.7	8.6	9.6	9.0	8.8
Precipitation	PrecAvg	18.8	1.2	0.9	1.0	0.8	1.9	2.8	3.0	2.4	1.8	1.1	1.1	1.1
	PrecMax	23.7	3.0	2.5	2.1	2.0	4.4	5.8	6.6	5.4	5.1	2.6	2.3	3.7
	PrecMin	13.0	0.2	0.2	0.0	0.2	0.2	0.6	0.6	0.3	0.2	0.2	0.1	0.2
	PrecStd	3.0	0.7	0.6	0.6	0.6	1.1	1.3	1.7	1.2	1.2	0.6	0.6	0.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	44.4	44.9	51.9	66.0	77.4	80.4	84.5	83.2	77.3	65.2	49.2	45.0
		MCWB	36.4	37.1	41.3	48.8	56.1	61.0	63.6	62.2	58.4	50.7	41.3	38.2
	2%	DB	41.1	40.8	46.4	59.9	72.3	75.9	79.6	78.4	72.2	58.2	44.0	41.2
		MCWB	35.4	34.4	38.1	45.3	53.2	58.1	61.8	61.5	56.6	46.8	38.0	35.4
	5%	DB	38.3	38.0	43.2	55.3	68.0	72.6	76.1	74.9	67.4	54.0	40.6	37.9
		MCWB	33.4	32.2	35.7	42.5	51.3	56.7	60.4	59.8	54.2	44.7	35.2	33.4
	10%	DB	35.0	35.0	40.3	51.5	64.3	69.3	72.9	71.4	63.4	50.6	37.7	34.0
		MCWB	30.8	30.0	33.7	40.2	49.4	54.9	59.0	58.2	52.0	42.3	33.1	30.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	38.2	37.9	41.9	49.8	58.7	63.5	66.5	66.3	60.6	52.6	41.9	38.7
		MCDB	43.5	43.7	50.1	64.2	72.1	76.5	78.2	76.3	73.6	63.3	48.2	44.0
	2%	WB	35.8	34.9	38.5	46.1	55.1	60.2	63.8	63.4	57.4	48.1	38.5	35.7
		MCDB	40.3	40.2	45.7	58.3	68.6	72.4	75.7	74.8	69.8	57.0	43.6	40.6
	5%	WB	33.7	32.6	36.1	43.2	52.5	58.1	61.7	61.1	54.9	45.2	35.6	33.4
		MCDB	38.0	37.6	42.5	53.5	65.3	69.7	73.4	72.6	65.7	52.9	39.6	37.8
	10%	WB	30.8	30.1	34.0	41.0	50.4	56.1	59.9	59.1	52.9	42.8	33.4	30.1
		MCDB	35.1	35.0	39.9	50.5	62.4	66.9	70.8	69.2	62.1	50.0	37.4	33.8
Mean Daily Temperature Range	5% DB	MDBR	12.9	13.9	15.0	17.0	19.3	18.2	18.3	19.0	17.6	13.3	11.4	12.4
		MCDBR	13.0	15.3	16.2	23.0	24.9	23.5	23.8	24.5	23.8	18.6	13.0	13.3
		MCWBR	10.0	11.3	10.7	12.9	12.4	10.9	10.9	11.1	12.2	11.0	9.7	10.6
	5% WB	MCDBR	13.7	15.2	16.4	21.7	23.3	21.7	22.5	22.8	22.6	18.2	12.9	13.9
		MCWBR	11.0	11.5	11.4	12.6	12.1	10.7	11.0	11.3	12.2	11.5	10.1	11.5
Clear-Sky Solar Irradiance	taub		0.233	0.248	0.274	0.318	0.344	0.344	0.344	0.341	0.307	0.282	0.239	0.218
	taud		2.301	2.351	2.330	2.376	2.404	2.408	2.446	2.451	2.548	2.569	2.404	2.242
	Ebn at Noon		226	269	286	287	284	285	283	276	273	255	224	199
	Edh at Noon		15	22	29	33	34	34	32	30	24	18	14	12
All-Sky Solar Radiation	RadAvg		204	490	946	1481	1740	1914	1853	1504	1023	535	241	147
	RadStd		21	40	63	68	119	172	106	132	93	58	23	10

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (13 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+113	N/A

Nomenclature: See separate page